High Speed Miniaturized Multiphoton Microscopy with Elliptical Beam Excitation

Ben Mattison[1,3], Shing-Jiuan Liu[2], Feng Tian[2], Weijian Yang[2,3,*]
[1]*Department of Biomedical Engineering, University of California, Davis, Davis, CA 95616, USA*
[2]*Department of Electrical and Computer Engineering, University of California, Davis, Davis, CA 95616, USA*
[3]*Biomedical Engineering Graduate Group, University of California, Davis, Davis, CA 95616, USA*
**wejyang@ucdavis.edu*

Abstract: We demonstrate elliptical beam shaping in a miniaturized two-photon microscope enabling cellular resolution fluorescence imaging at over three times faster frame rate than point scanning over a 400 µm diameter FOV. © 2023 The Author(s)

1. Introduction

Miniaturized multiphoton microscopes which image cellular resolution neural activity in freely moving animals are promising tools in neuroscience with many exciting new advances [1-4]. However, multiphoton imaging traditionally suffers from imaging speed limitations due to the necessity of scanning the laser excitation spot over the entire field of view (FOV) to generate a single image frame. There is therefore an intrinsic trade-off between a well-sampled large FOV and the overall imaging frame rate for a given fixed scanner speed. Typically, a raster scanning pattern is used, and the frame rate is determined by the single-line scanning speed (set by the scanner properties) divided by the number of lines in the raster. Beam shaping and point-spread-function (PSF) engineering can overcome these limitations by forgoing a diffraction limited spot and adapting the laser excitation spot to match the morphology of the sample targets and thus possibly reducing the number of lines required for a single frame [5,6]. Such a strategy is particularly suitable for functional imaging of neural population activity in rodents using calcium or voltage indicators, where the imaging targets are the neuronal cell bodies with a typical diameter of ~10-15 µm. Therefore, an elliptical beam shape with a long axis on the order of ~5-10 µm could still image similar neural information as a point scanning method but could greatly reduce the number of scanning lines per frame. Here, for the first time, we demonstrated elliptical beam shaping in a miniaturized two-photon microscope through an innovative optical setup, and we increased the frame rate by five times compared to a point scanning scheme.

2. Miniaturized Elliptical Beam Shaper

A miniaturized elliptical beam shaper (MEBS) using a two-staged cylindrical lens fiber collimation was developed (Fig. 1). A hollow-core photonic crystal fiber (HC-PhC) fiber is held by a ceramic ferrule and adjusted relative to two cylindrical collimating lenses, CL1 and CL2, using 3D printed holders. CL2 collimates the beam in the X-axis (Fig. 1a), and CL1 collimates the beam in the Y-axis (Fig. 1b). The MEBS output is a collimated beam of ~1.1 mm × 0.12 mm FWHM (Fig. 1c-d). We designed such a beam profile so the elliptical shape PSF at the sample plane could still be tightly confined axially.

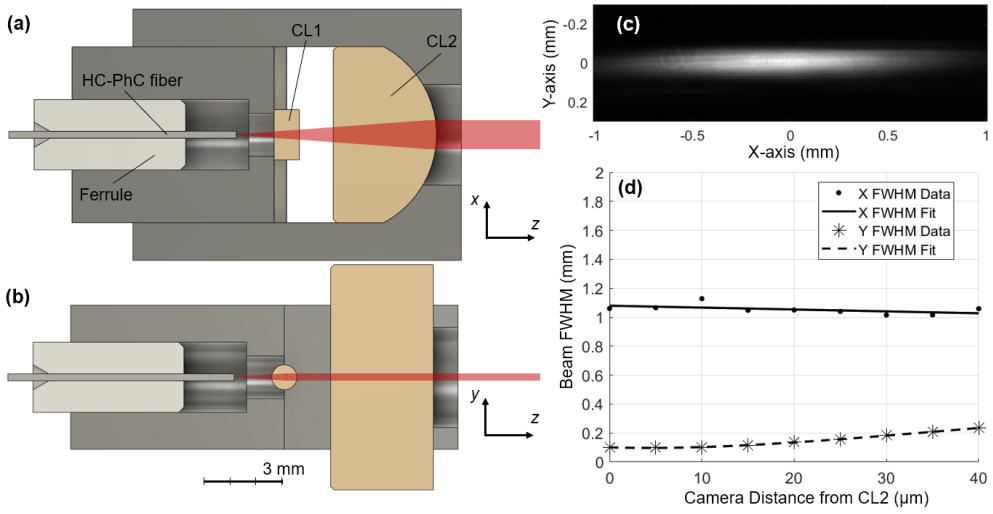


Fig. 1. Setup of the miniaturized elliptical beam shaper (MEBS) diagram in (a) X-Z and (b) Y-Z planes. (c) CMOS camera image of elliptical beam output in the transverse plane. (d) Measured beam profile after the MEBS, showing the low beam divergence in each axis.

The output from the MEBS is used as a collimated beam input to a modified version of an open-source two-photon miniscope design, MINI2P [2]. The beam is scanned using an off-the-shelf microelectromechanical (MEMS) scanner and the laser excitation and fluorescence collection occurs through a commercially available objective lens. Fluorescence from the sample passes through a dichroic mirror and is collected using a second optical fiber.

3. Results

We measured the PSF by imaging the excitation beam profile in a fluorescein cuvette through a side-view camera (Fig. 2). The two-photon fluorescence PSF was measured to be ~1.6 µm × ~5.7 µm FWHM laterally and ~18 µm FWHM axially with consistent performance across a 400 µm FOV.

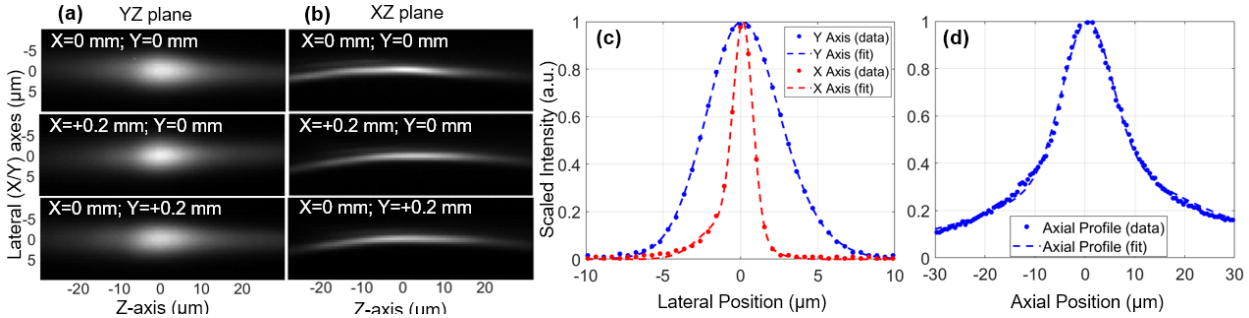


Fig. 2. PSF measurement through a fluorescein cuvette with side-view camera. Ellipitical PSF profiles in (a) YZ and (b) XZ planes are shown at the center and edge of the FOV. (c) Lateral profiles of the elliptical PSF's short (X) and long (Y) axes. (d) Axial (Z) profile of PSF.

Proof-of-principle fluorescence imaging was done with 5 µm fluorescent beads and pollen grains to compare the results of standard point scanning versus the increased frame rate elliptical beam scanning method (Fig. 3). Point scanning was acquired by scanning the ~400 µm FOV with 256×256 pixels (Fig. 3a, c). Elliptical beam scanning was acquired with a 256×80 pixel scan (Fig. 3b, d). Importantly, when comparing the same FOV, beads and pollen grains visible in point scanning are also visible with elliptical beam scanning, indicating how neuronal cell bodies could be similarly extracted with no loss of neural signal between elliptical and point scanning schemes. Because the elliptical beam scanning method (Fig. 3b, d) uses 3.2× fewer lines, the frame rate acquisition is 3.2× faster than the point scanning. The MEBS enables a significant increase in frame rate when imaging neuronal activity in freely moving animals. Ongoing work includes optimization of the excitation beam profile to further increase the frame rate while maintaining cellular resolution and demonstration of high-speed imaging of neural activity in *in vivo* mouse brain.

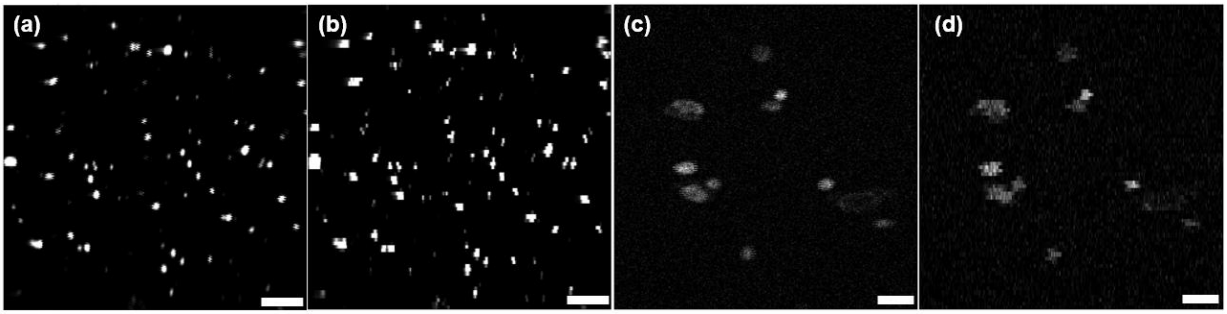


Fig. 3. Comparison of fluorescence imaging of 5 µm fluorescent beads (a)-(b) and pollen grains (c)-(d) between point scanning (a, c) and our elliptical beam scanning scheme (b, d). Scale bar: (a-b) 50 µm, (c-d) 50 µm.

4. Acknowledgement

We acknowledge support from National Institute of Neurological Disorders and Stroke and National Eye Institute (R01NS118289), and Burroughs Wellcome Fund (Career Award at the Scientific Interface 1015761).

5. References

[1] Guo, Changliang, et al. "New imaging instrument in animal models: Two-photon miniature microscope and large field of view miniature microscope for freely behaving animals." *Journal of Neurochemistry* (2022).
[2] Zong, Weijian, et al. "Large-scale two-photon calcium imaging in freely moving mice." *Cell* 185.7 (2022): 1240-1256.
[3] Klioutchnikov, Alexandr, et al. "A three-photon head-mounted microscope for imaging all layers of visual cortex in freely moving mice." *Nature Methods* (2022): 1-7.
[4] Zhao, Chunzhu, et al. "Miniature three-photon microscopy maximized for scattered fluorescence collection." *Nature Methods* (2023): 1-6.
[5] Prevedel, Robert, et al. "Fast volumetric calcium imaging across multiple cortical layers using sculpted light." *Nature methods* 13.12 (2016): 1021-1028.
[6] Weisenburger, Siegfried, et al. "Volumetric Ca2+ imaging in the mouse brain using hybrid multiplexed sculpted light microscopy." *Cell* 177.4 (2019): 1050-1066.